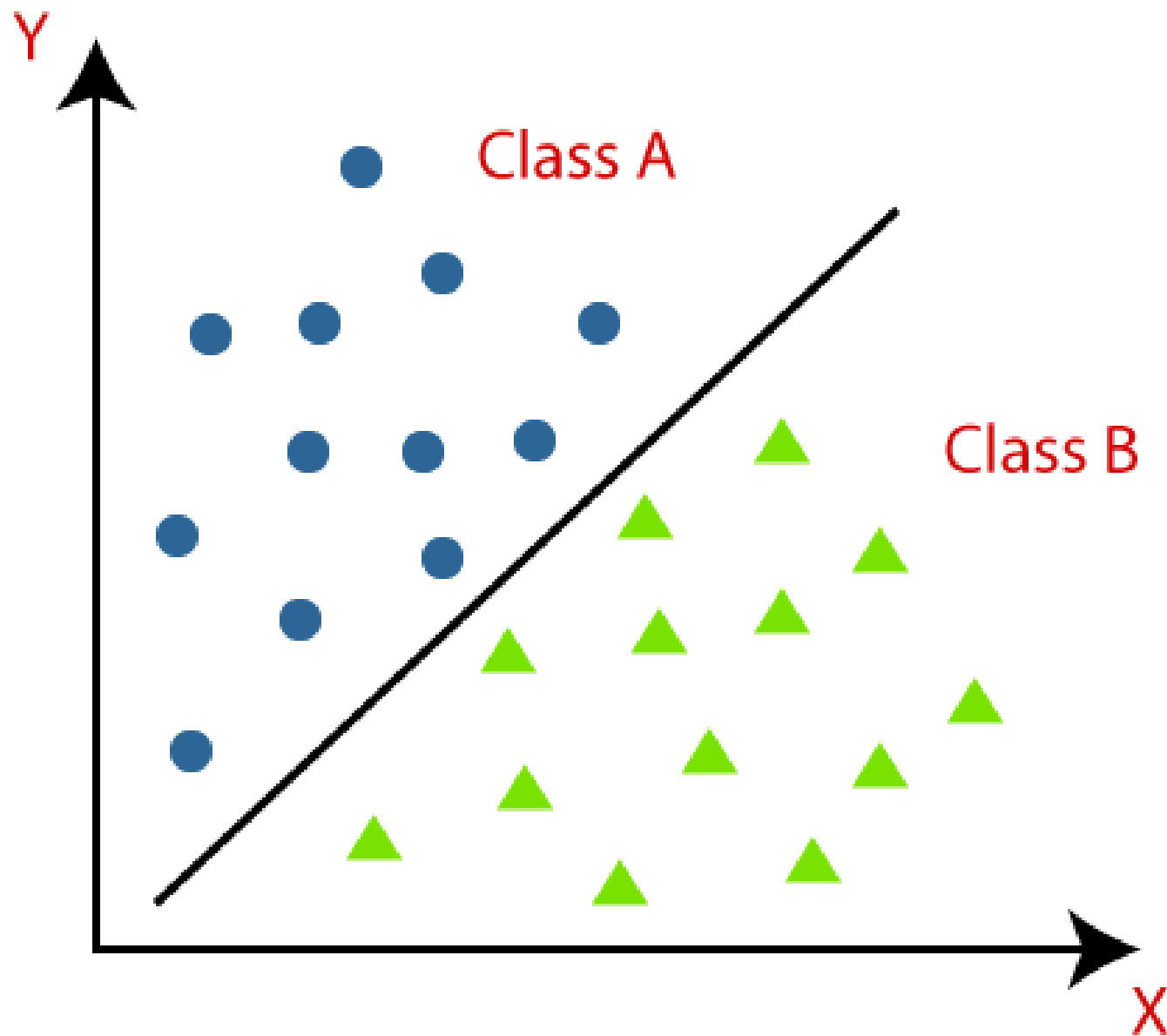


Classification





Reading Assignment

Chapter 8 of
Introduction to Statistical Learning
By Gareth James, et al.

Classification: Definition

- Given a collection of records (training set)
 - Each record is by characterized by a tuple (x,y) , where x is the attribute set and y is the class label
 - ◆ x : attribute, predictor, independent variable, input
 - ◆ y : class, response, dependent variable, output
- Task:
 - Learn a model that maps each attribute set x into one of the predefined class labels y

Examples of Classification Task

Task	Attribute set, x	Class label, y
Categorizing email messages	Features extracted from email message header and content	spam or non-spam
Identifying tumor cells	Features extracted from MRI scans	malignant or benign cells
Cataloging galaxies	Features extracted from telescope images	Elliptical, spiral, or irregular-shaped galaxies

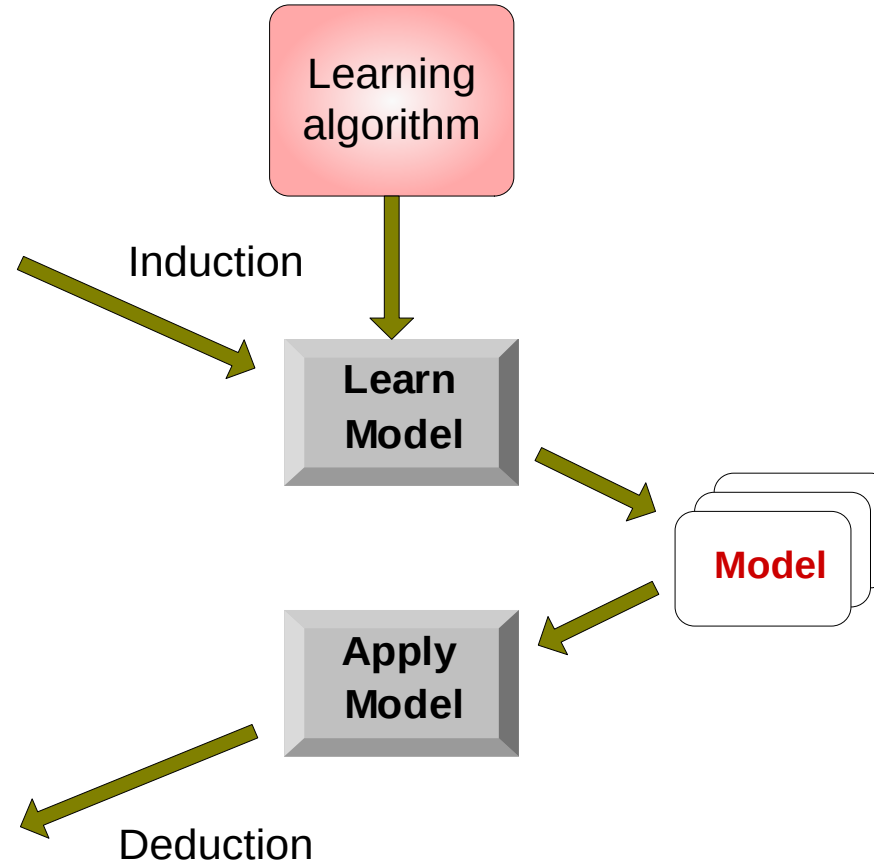
General Approach for Building Classification Model

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
9	No	Medium	75K	No
10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
12	Yes	Medium	80K	?
13	Yes	Large	110K	?
14	No	Small	95K	?
15	No	Large	67K	?

Test Set



Classification Techniques

□ Base Classifiers

- Decision Tree based Methods
- Rule-based Methods
- Nearest-neighbor
- Neural Networks
- Deep Learning
- Naïve Bayes and Bayesian Belief Networks
- Support Vector Machines

□ Ensemble Classifiers

- Boosting, Bagging, Random Forests

A decision tree is a classifier in the form of a tree structure with two types of nodes:

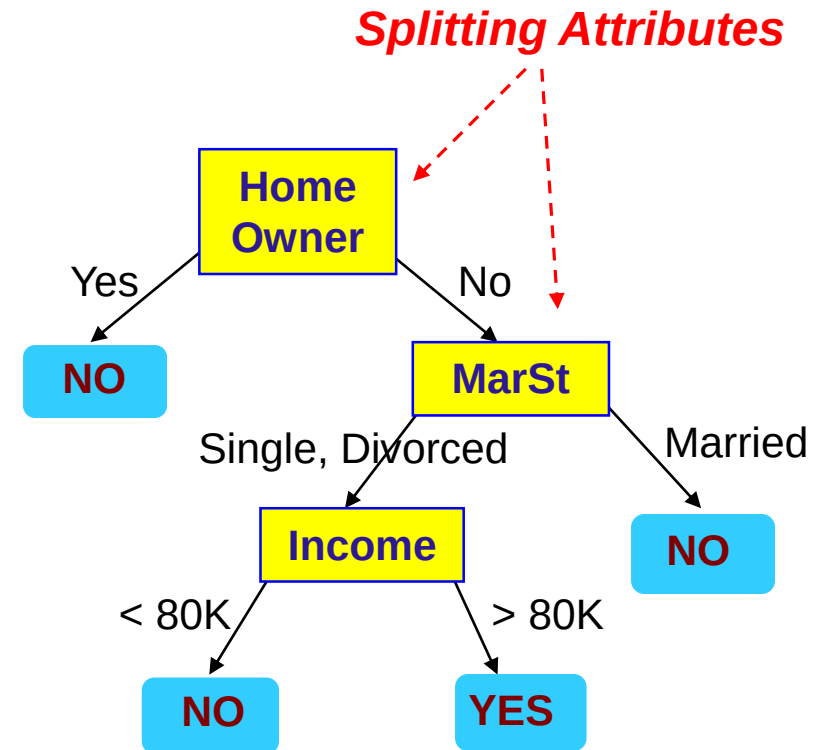
- Decision node: Specifies a choice or test of some attribute, with one branch for each outcome
- Leaf node: Indicates classification of an example

Example of a Decision Tree

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

categorical
categorical
continuous
class

Training Data

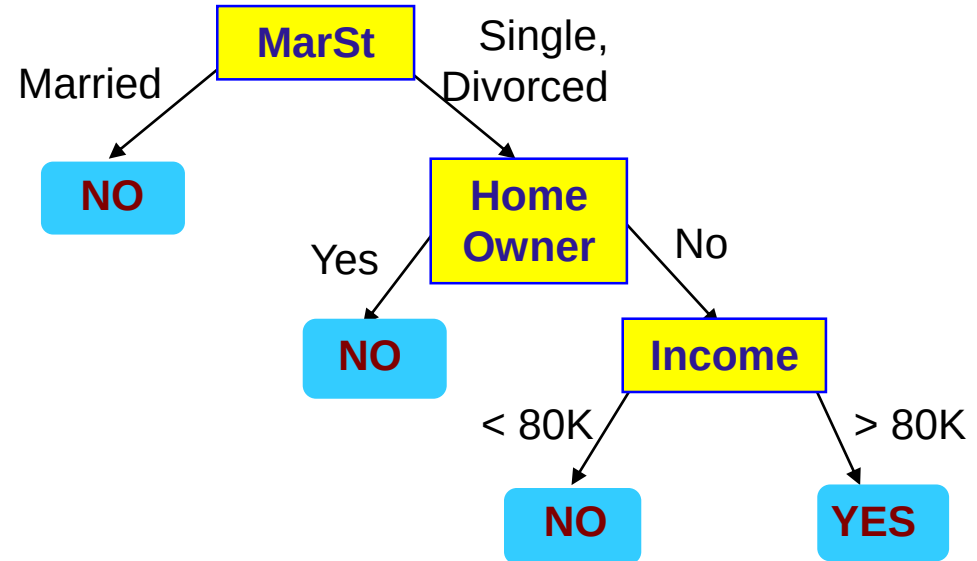


Model: Decision Tree

Another Example of Decision Tree

ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

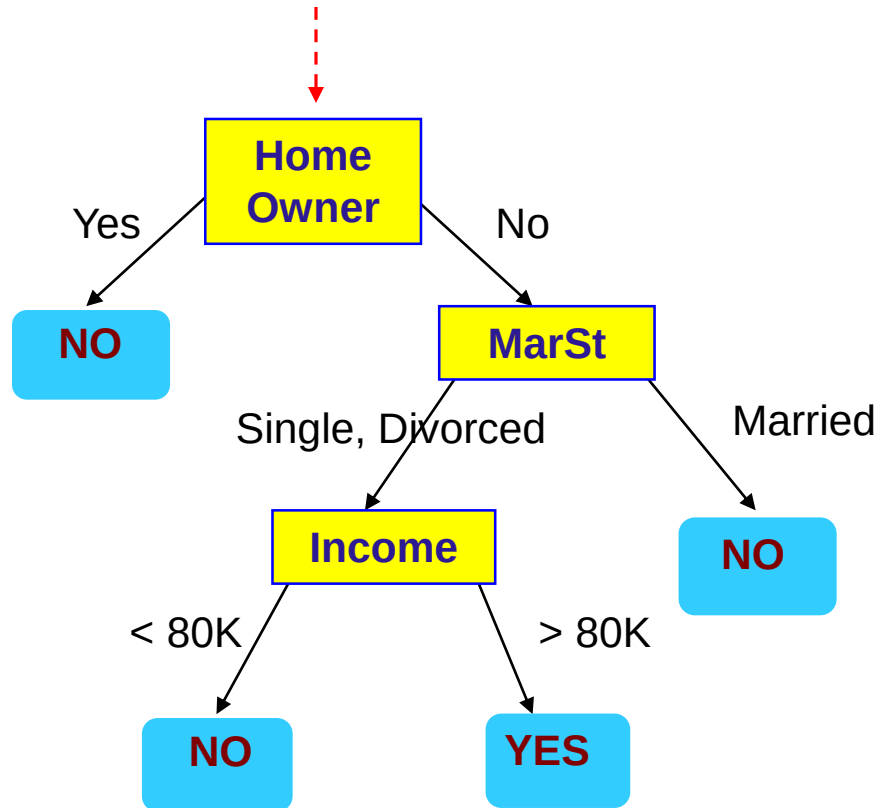
categorical
categorical
continuous
class



There could be more than one tree that fits the same data!

Apply Model to Test Data

Start from the root of tree.



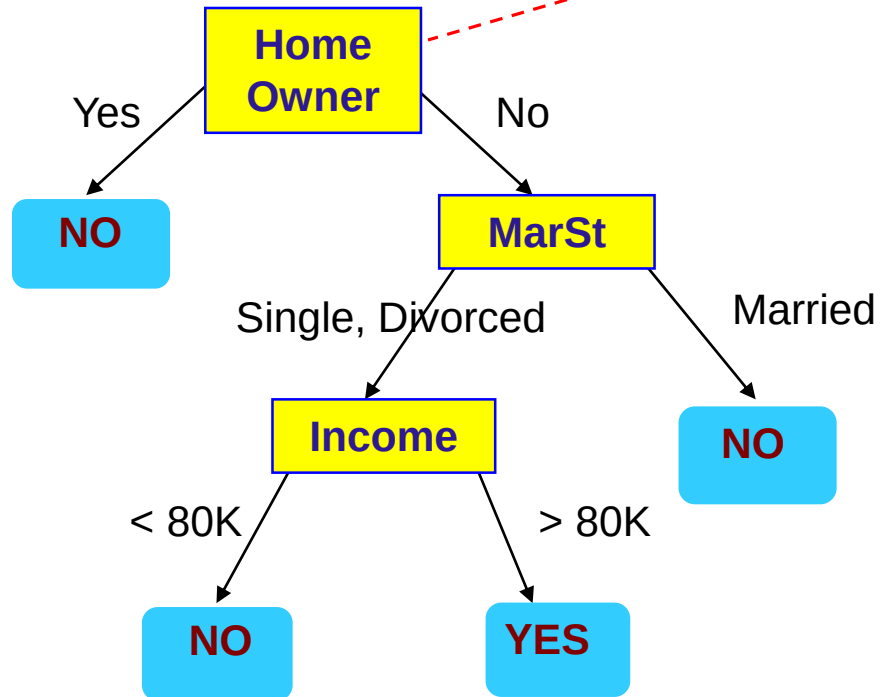
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?

Apply Model to Test Data

Test Data

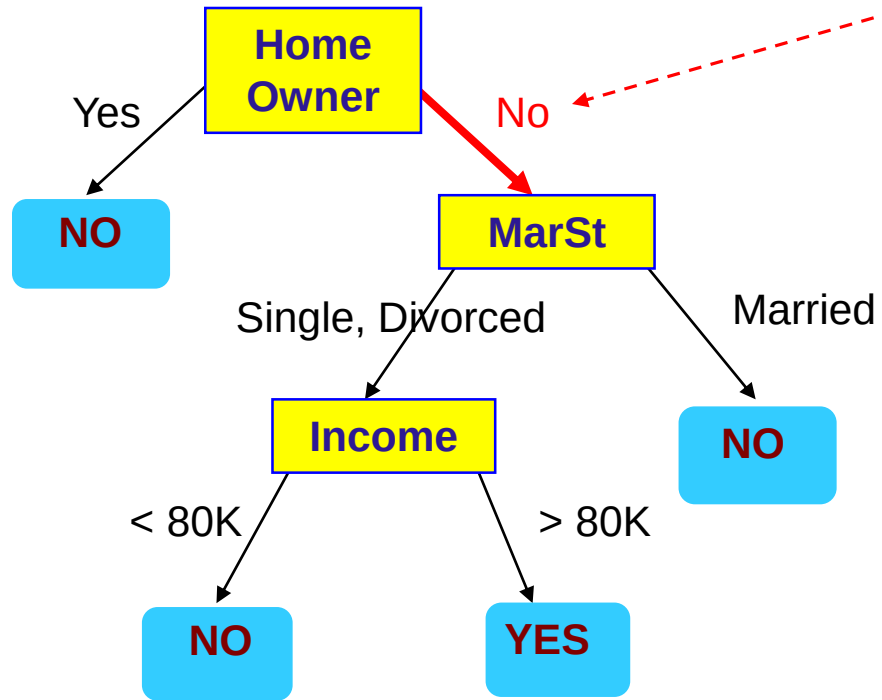
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

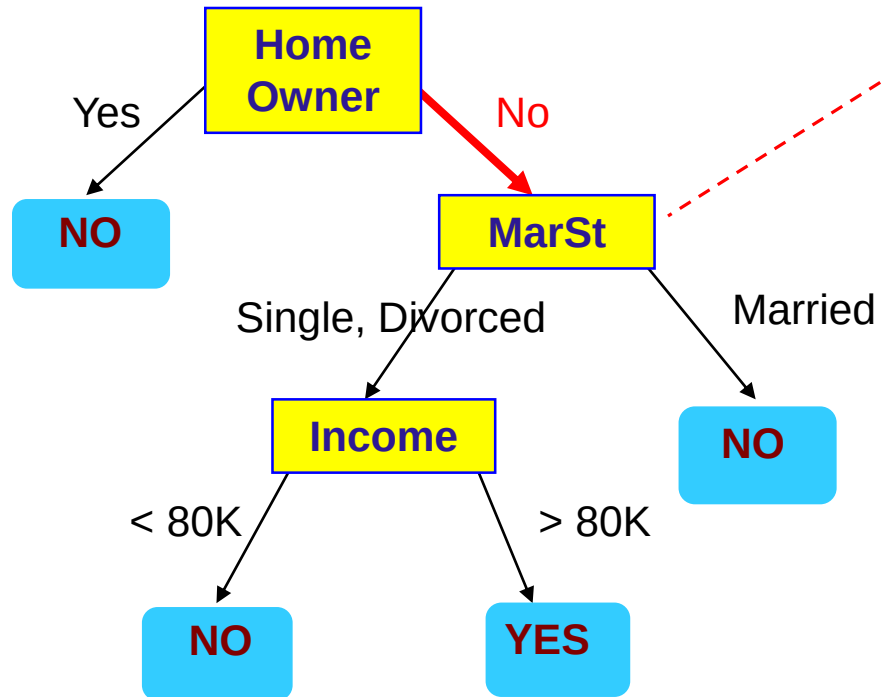
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

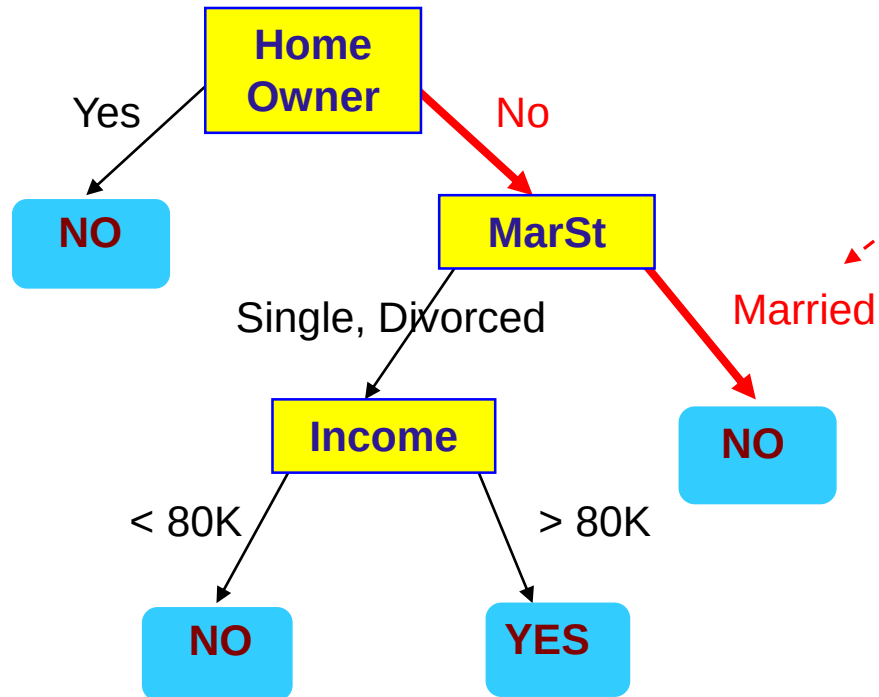
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

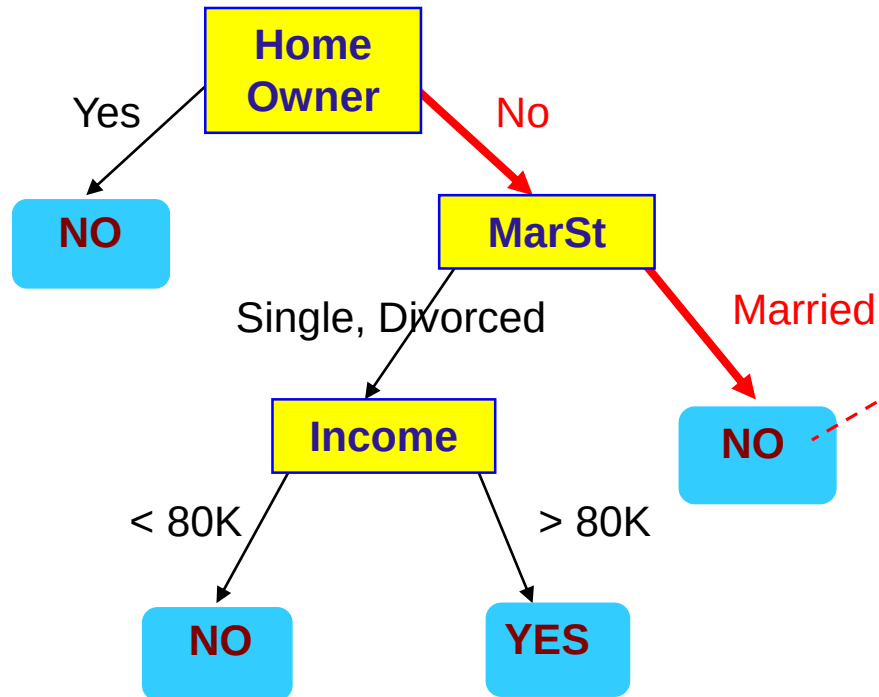
Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Apply Model to Test Data

Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Assign Defaulted to
"No"

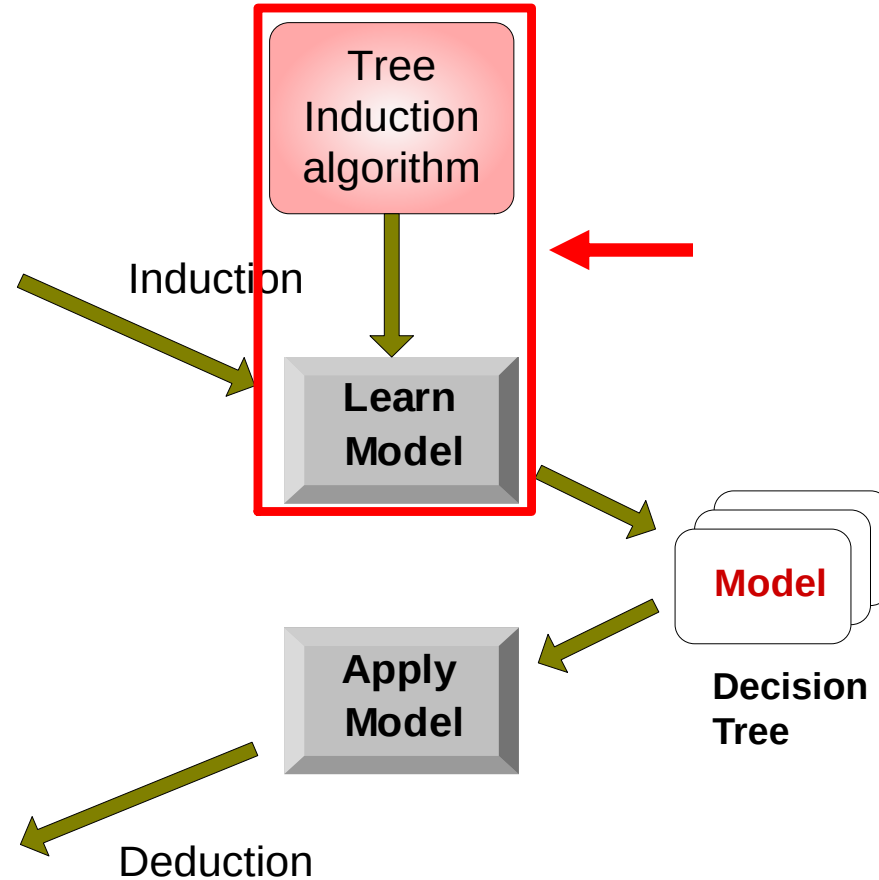
Decision Tree Classification Task

Tid	Attrib1	Attrib2	Attrib3	Class
1	Yes	Large	125K	No
2	No	Medium	100K	No
3	No	Small	70K	No
4	Yes	Medium	120K	No
5	No	Large	95K	Yes
6	No	Medium	60K	No
7	Yes	Large	220K	No
8	No	Small	85K	Yes
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10	No	Small	90K	Yes

Training Set

Tid	Attrib1	Attrib2	Attrib3	Class
11	No	Small	55K	?
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14	No	Small	95K	?
15	No	Large	67K	?

Test Set

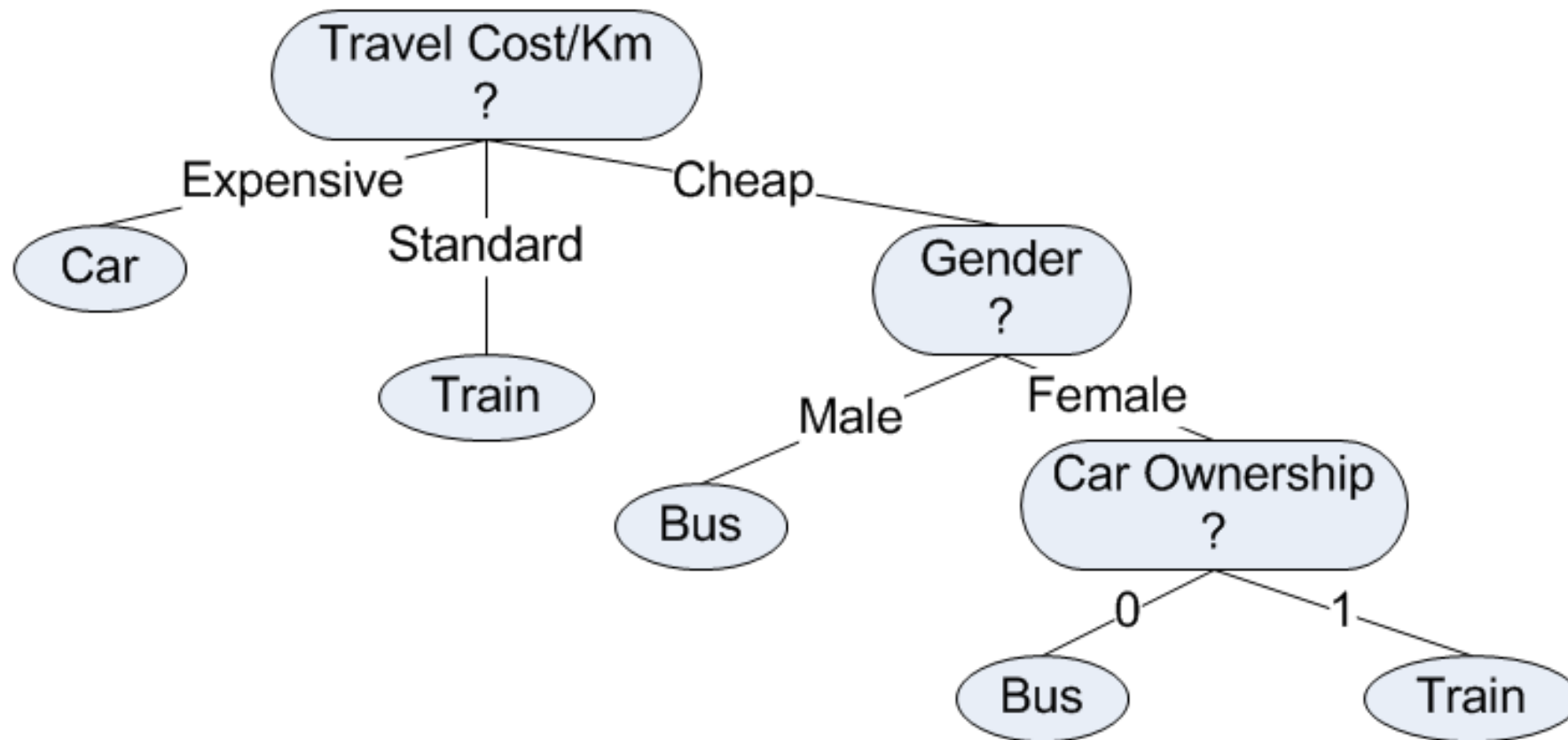


Example Classification Task

Attributes				Classes
Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Based on above training data, we can induce a decision tree as the following:

Example Classification Task



what transportation mode would Alex, Buddy and Cheery use?

Person name	Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation Mode
Alex	Male	1	Standard	High	?
Buddy	Male	0	Cheap	Medium	?
Cherry	Female	1	Cheap	High	?

what transportation mode would Alex, Buddy and Cheery use?

- Based on the rules or decision tree above, the classification is very straightforward.
- Alex is willing to pay standard travel cost per km, thus regardless his other attributes, his transportation mode must be train.
- Buddy is only willing to pay cheap travel cost per km, and his gender is male, thus his selection of transportation mode should be bus.
- Cherry is also willing to pay cheap travel cost per km, and her gender is female and she owns a car, thus her transportation mode choice to work is train (probably she uses car only during weekend to shop).
- Variable Income level never be utilized to classify the transportation mode in this case.

Issues with the Classification Example

- First, it must be noted, however, that with limited number of training data (only 10) that induce the decision tree, we cannot generalize the rules of the decision tree above to be applicable for other cases in the city.
- The sequence of rules generated by the decision tree is based on priority of the attributes. For example, there is no rule for people who own more than 1 car because based on the data it is already covered by attribute travel cost/km. For those who own 2 cars the travel cost/km are always expensive, thus the mode is car.

Issues with the Decision Trees

Due to the limitation of decision algorithm (most algorithms of decision tree employ greedy strategy with no backtracking thus it is not exhaustive search), these sequences of priority in general is not optimum. We cannot say that the rules generated by decision tree are the best rules.



Intuition Behind Splits

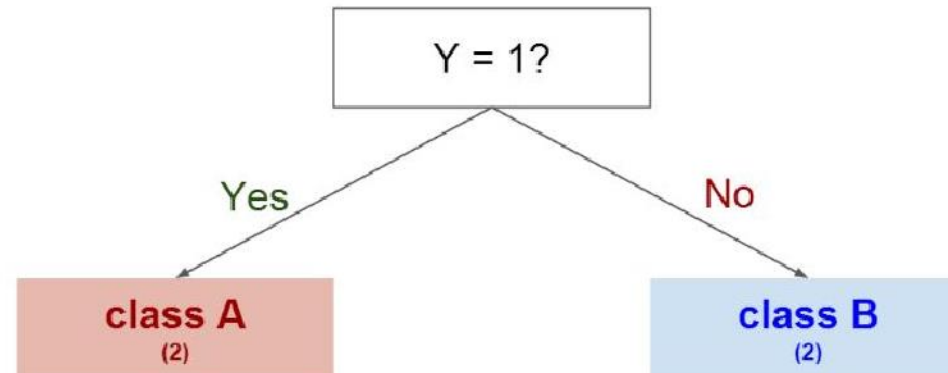
Imaginary Data with 3 features (X,Y, and Z) with two possible classes.

X	Y	Z	Class
1	1	1	A
1	1	0	A
0	0	1	B
1	0	0	B



Intuition Behind Splits

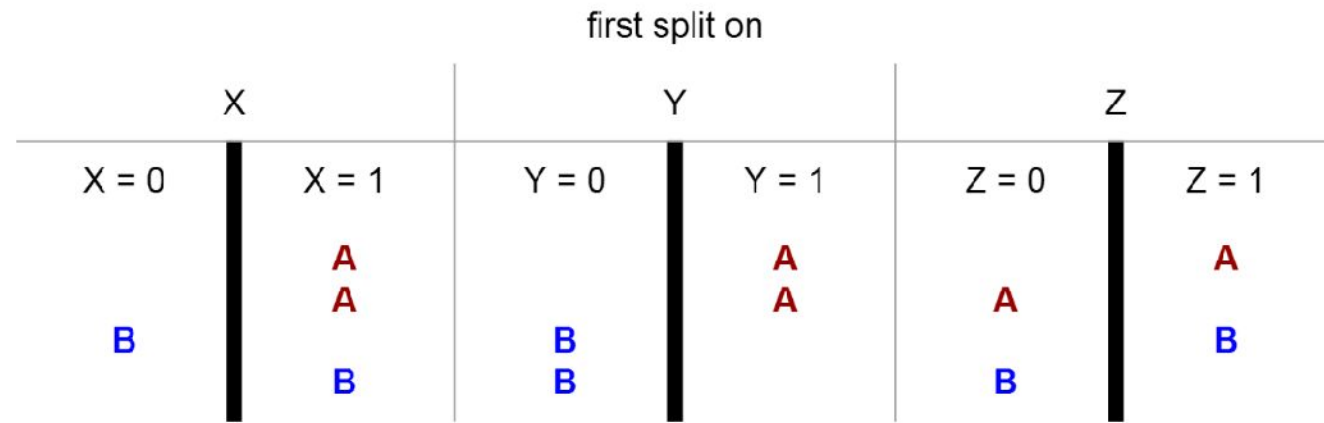
Splitting on Y gives us a clear separation between classes





Intuition Behind Splits

We could have also tried splitting on other features first:



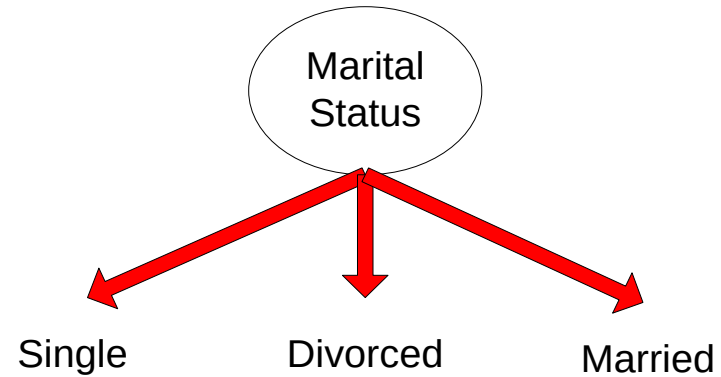
Methods for Expressing Test Conditions

- | Depends on attribute types
 - Binary
 - Nominal
 - Ordinal
 - Continuous
- | Depends on number of ways to split
 - 2-way split
 - Multi-way split

Test Condition for Nominal Attributes

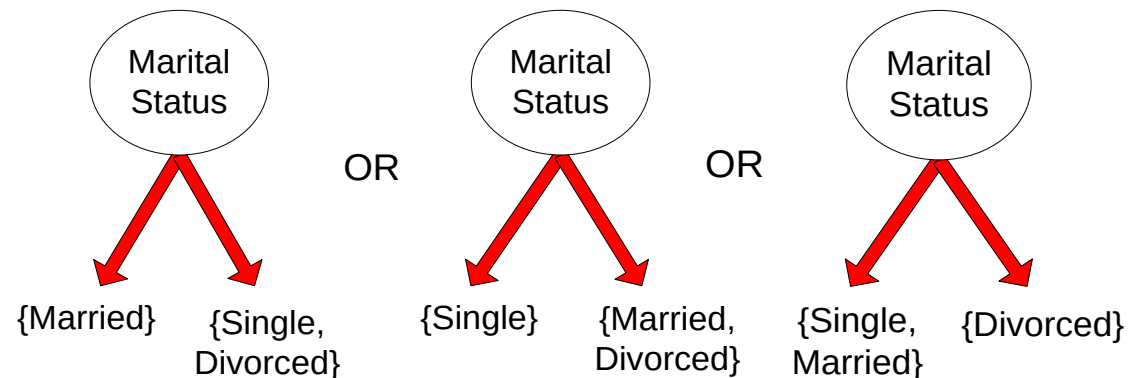
- **Multi-way split:**

- Use as many partitions as distinct values.



- **Binary split:**

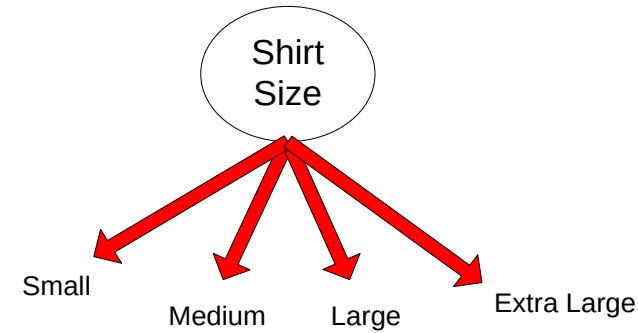
- Divides values into two subsets



Test Condition for Ordinal Attributes

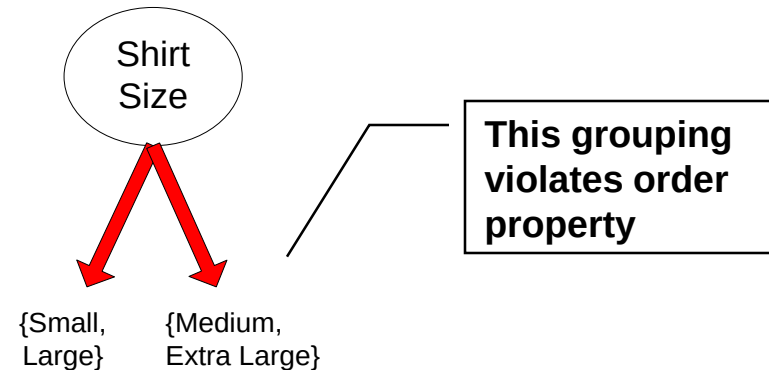
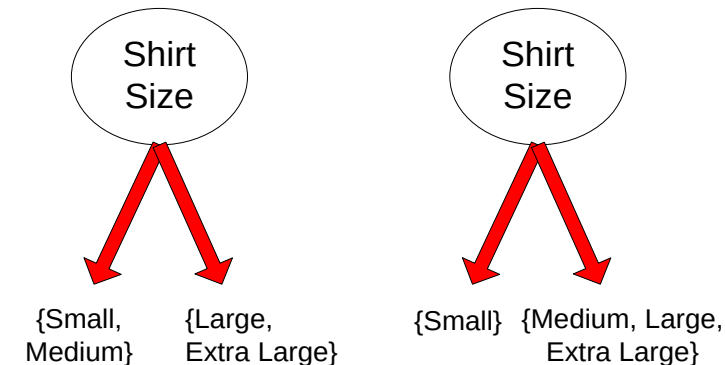
| Multi-way split:

- Use as many partitions as distinct values

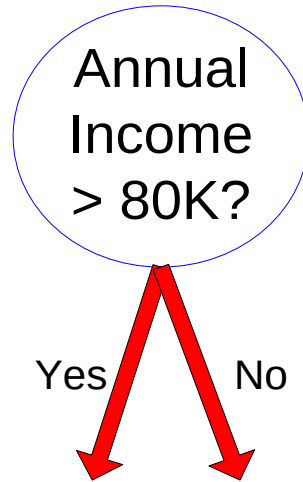


| Binary split:

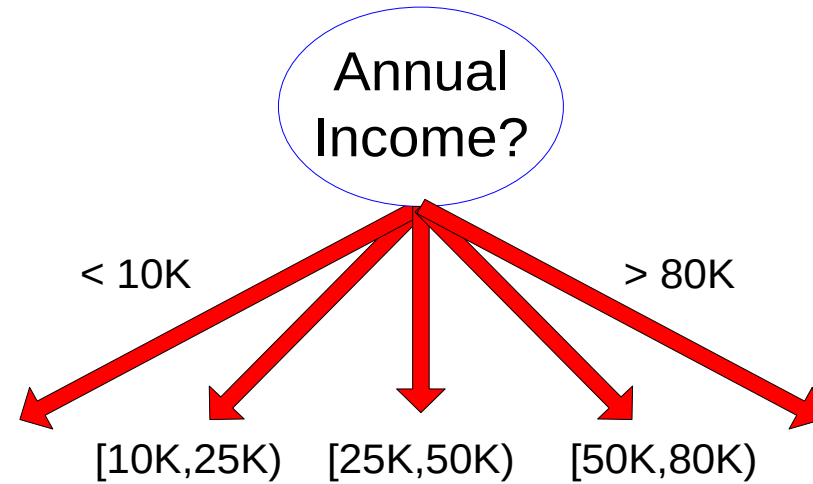
- Divides values into two subsets
- Preserve order property among attribute values



Test Condition for Continuous Attributes



(i) Binary split



(ii) Multi-way split

Splitting Based on Continuous Attributes

- Different ways of handling

- **Discretization** to form an ordinal categorical attribute

- Ranges can be found by equal interval bucketing, equal frequency bucketing (percentiles), or clustering.

- ◆ Static – discretize once at the beginning
 - ◆ Dynamic – repeat at each node

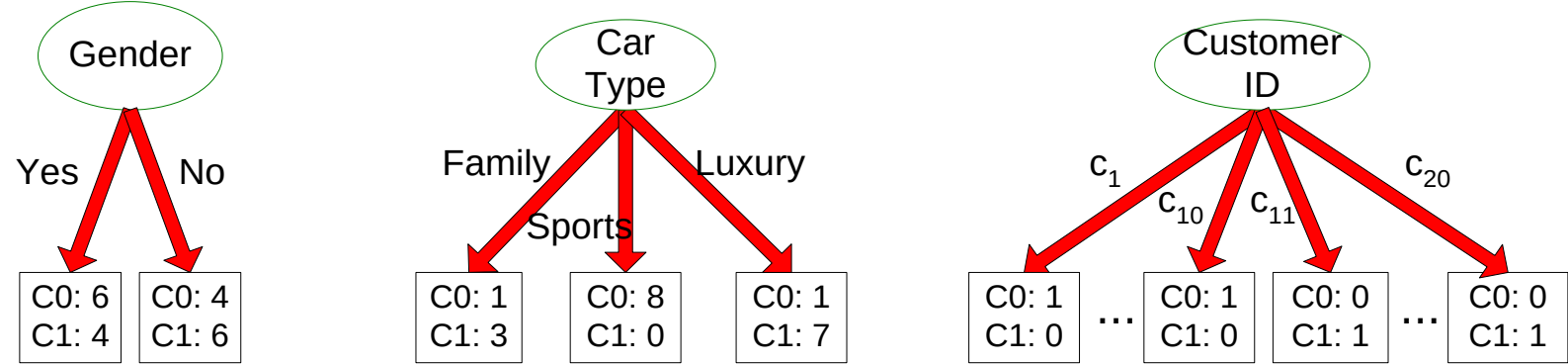
- **Binary Decision**: $(A < v)$ or $(A \geq v)$

- ◆ consider all possible splits and finds the best cut
 - ◆ can be more compute intensive

How to determine the Best Split

Before Splitting: 10 records of class 0,
10 records of class 1

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1



Which test condition is the best?

How to measure impurity?

- Given a data table that contains attributes and class of the attributes, we can measure homogeneity (or heterogeneity) of the table based on the classes.
- We say a table is pure or homogenous if it contains only a single class.
- If a data table contains several classes, then we say that the table is impure or heterogeneous.
- There are several indices to measure degree of impurity quantitatively. Most well known indices to measure degree of impurity are entropy, Gini index, and classification error.

Measures of Node Impurity

- | Gini Index

$$GINI(t) = 1 - \sum_j [p(j|t)]^2$$

- | Entropy

$$Entropy(t) = -\sum_j p(j|t) \log p(j|t)$$

- | Misclassification error

$$Error(t) = 1 - \max_i P(i|t)$$

Finding the Best Split

1. Compute impurity measure (P) before splitting
2. Compute impurity measure (M) after splitting
 - | Compute impurity measure of each child node
 - | M is the weighted impurity of children
3. Choose the attribute test condition that produces the highest gain

$$\text{Gain} = P - M$$

or equivalently, lowest impurity measure after splitting (M)

Calculation of Impurity in Classification

Example of Slides 17-20

In our example, the classes of Transportation mode below consist of three groups of Bus, Car and Train. In this case, we have 4 buses, 3 cars and 3 trains (in short we write as 4B, 3C, 3T). The total data is 10 rows.

Attributes				Classes
Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car
Female	1	Cheap	Medium	Train
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train

Based on these data, we can compute probability of each class. Since probability is equal to frequency relative, we have

$$\text{Prob (Bus)} = 4 / 10 = 0.4$$

$$\text{Prob (Car)} = 3 / 10 = 0.3$$

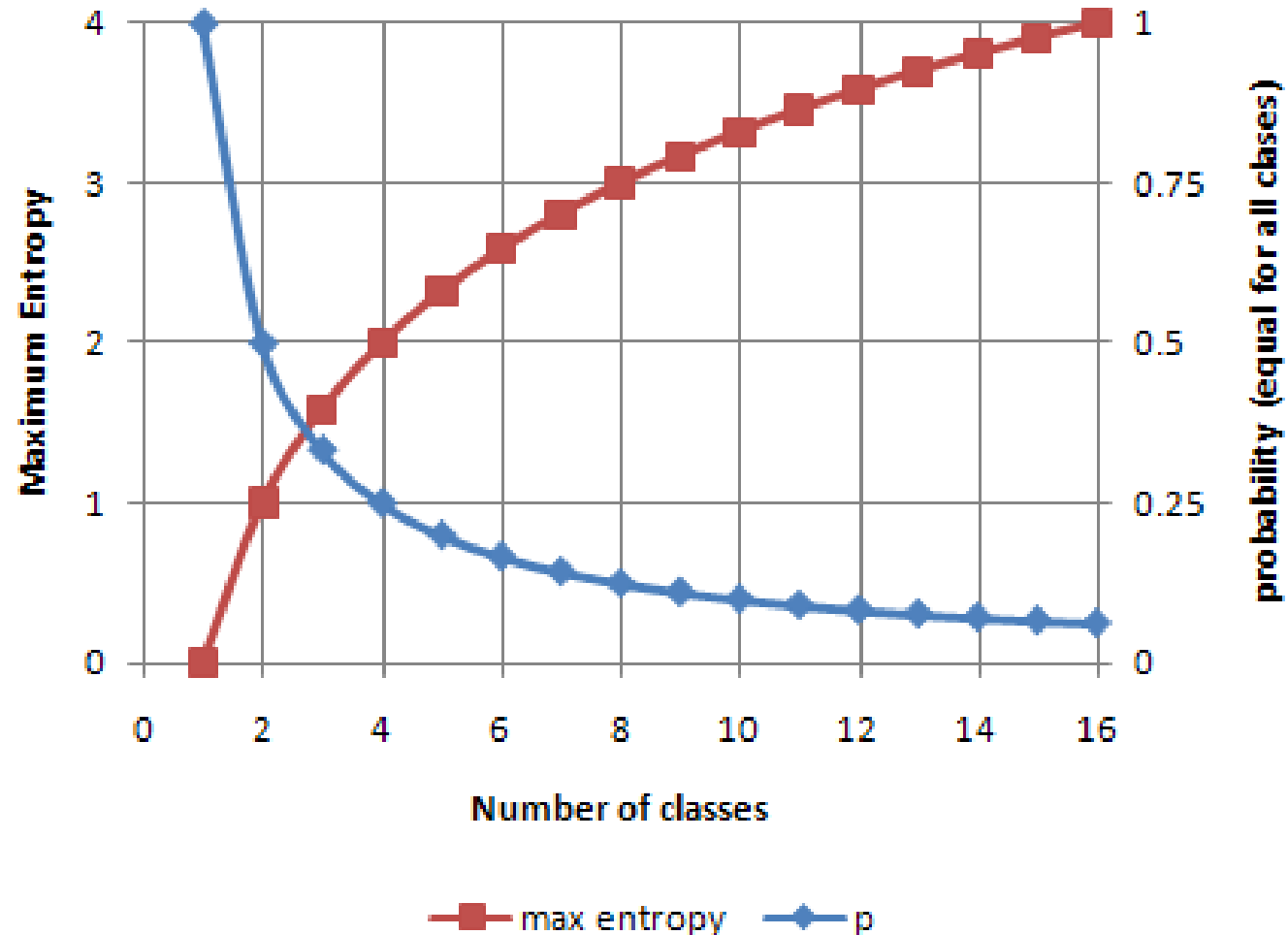
$$\text{Prob (Train)} = 3 / 10 = 0.3$$

Calculation of **Entropy** in Classification Example of Slides 17-20

- Example: Given that $\text{Prob}(\text{Bus}) = 0.4$, $\text{Prob}(\text{Car}) = 0.3$ and $\text{Prob}(\text{Train}) = 0.3$, we can now compute entropy as
- Entropy = $-0.4 \log(0.4) - 0.3 \log(0.3) - 0.3 \log(0.3) = 1.571$
- The logarithm is base 2.

Calculation of Entropy in Classification

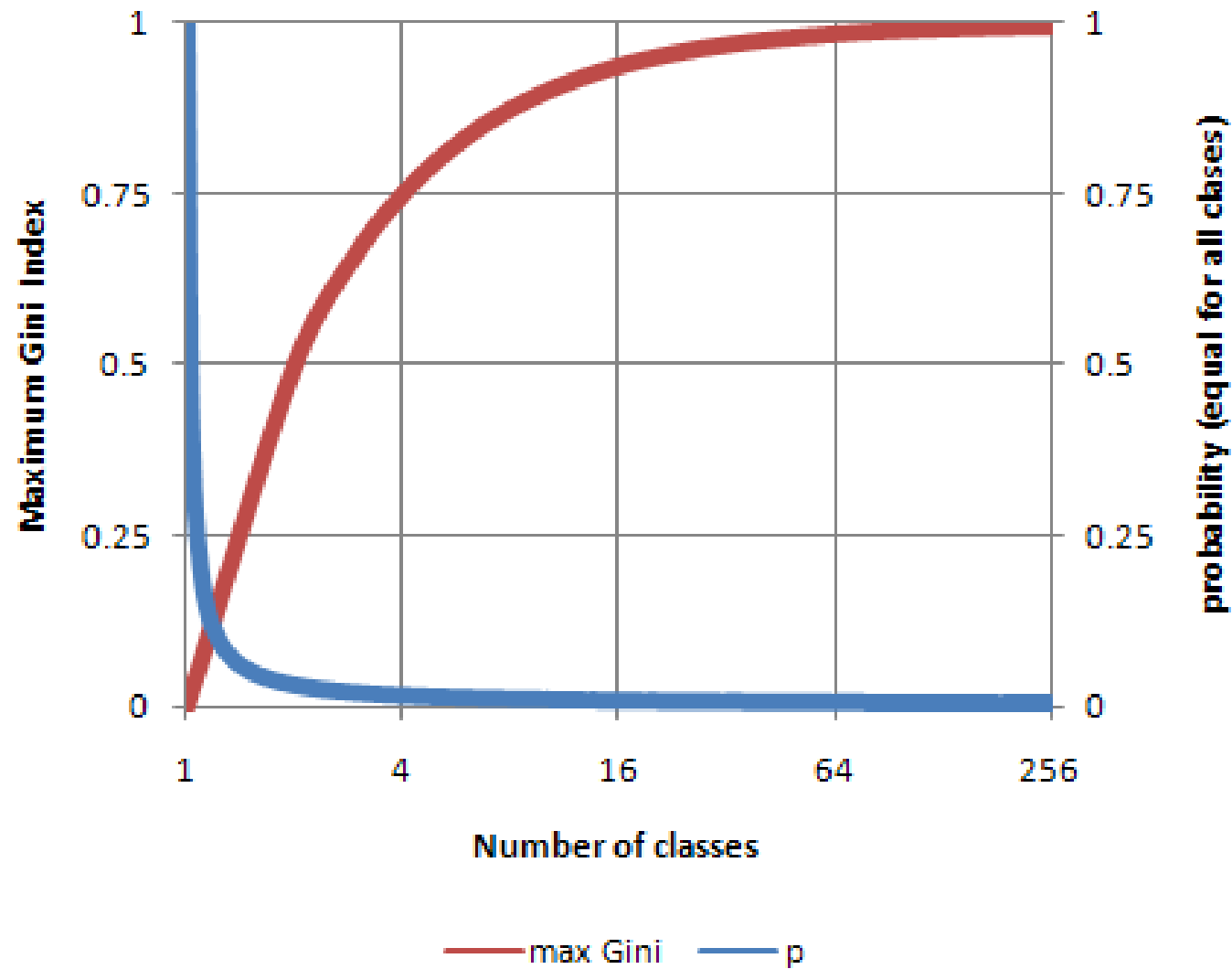
Example of Slides 17-20



Calculation of **Gini Index** in Classification Example of Slides 17-20

- Example: Given that Prob (Bus) = 0.4, Prob (Car) = 0.3 and Prob (Train) = 0.3, we can now compute Gini index as
- Gini Index = $1 - (0.4^2 + 0.3^2 + 0.3^2) = 0.660$

Calculation of Gini Index in Classification Example of Slides 17-20

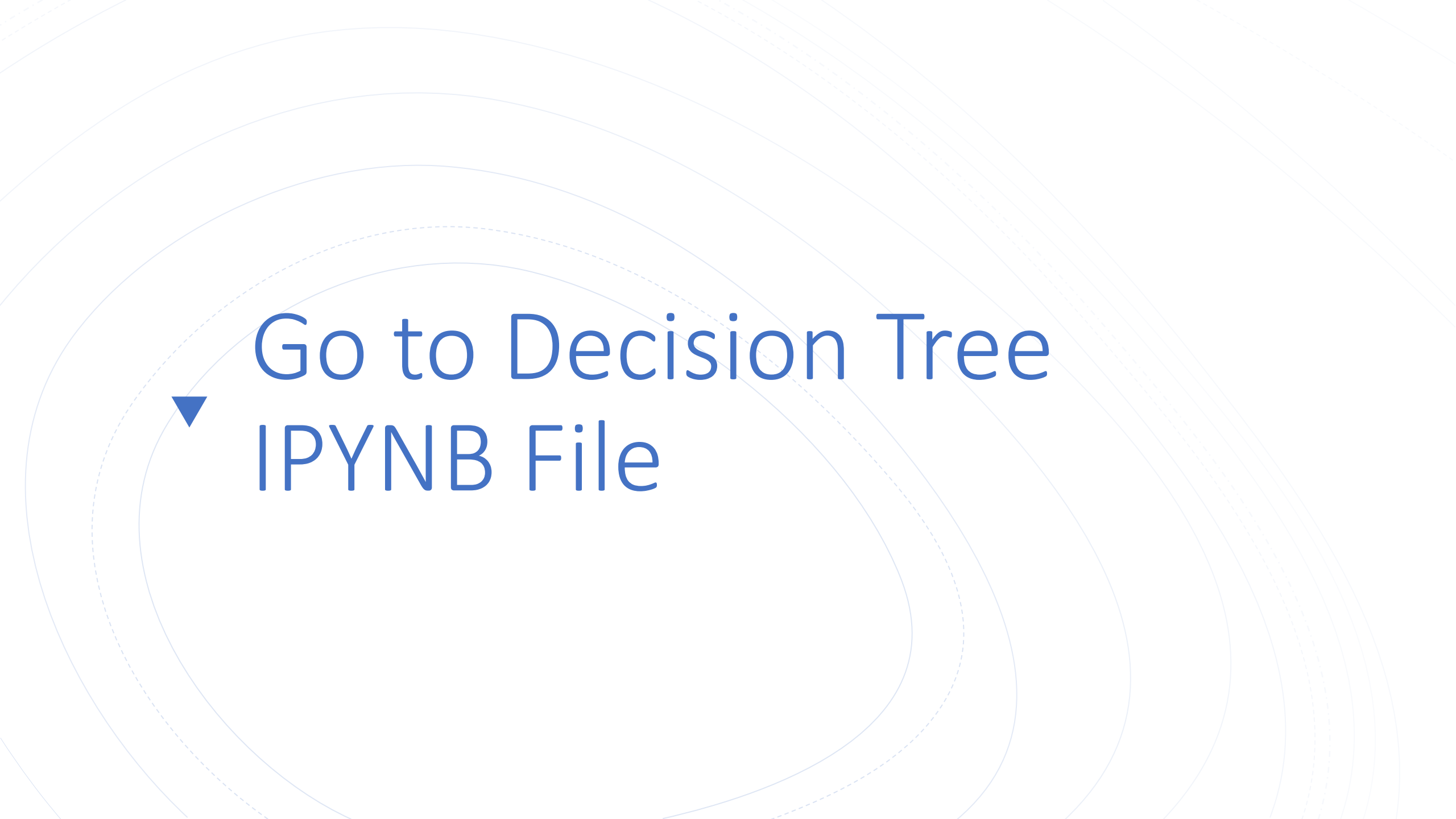


Calculation of **Classification error** in Classification Example of Slides 17-20

- Example: Given that Prob (Bus) = 0.4, Prob (Car) = 0.3 and Prob (Train) = 0.3, index of classification error is given as
- Classification Error Index = $1 - \text{Max}\{0.4, 0.3, 0.3\}$
 $= 1 - 0.4 = 0.60$

Decision Tree Induction

- Many Algorithms:
 - Hunt's Algorithm (one of the earliest, greedy algorithm and simplest to implement)
 - CART
 - ID3, C4.5
 - SLIQ, SPRINT



▼ Go to Decision Tree
IPYNB File